DAY 1

Foundations & Amplicon Metagenomics

What is metagenomics • Types of sequencing • 16S rRNA gene
Databases • Amplicon analysis pipeline

Today's Learning Objectives

01 Define metagenomics

Explain the concept of studying genetic material from environmental samples without culturing.

02 Compare sequencing strategies

Distinguish amplicon from shotgun metagenomics and choose the right approach.

03 Understand the 16S rRNA gene

Describe its structure, variable regions, and why it is used as a phylogenetic marker.

04 Navigate key databases

Identify SILVA, Greengenes2, NCBI RefSeq and their roles in taxonomy assignment.

05 Trace the amplicon pipeline

Follow raw reads from QC to taxonomy using QIIME2, DADA2 and EMU.

06 Hands-on with real data

Apply FASTQC, SEQKIT and QIIME2 to human gut and soil 16S datasets.

What is Metagenomics?

"The study of genetic material recovered directly from environmental samples, without prior cultivation of organisms."

Culture-independent

Study microbes that cannot be grown in the lab — >99% of environmental microbes are unculturable.

Community-level

Analyse entire microbial communities simultaneously: bacteria, archaea, viruses, fungi in one experiment.

High-throughput

Next-generation sequencing generates millions of reads, enabling deep coverage of complex communities.

Term coined by Jo Handelsman et al. (1998) studying soil communities.

Why Metagenomics? Applications Across Fields

Human Health

Gut microbiome & disease (IBD, obesity, T2D)
Probiotics & precision medicine
Hospital-acquired infection surveillance

Environmental Science

Soil health & agriculture
Ocean & freshwater ecology
Biodiversity monitoring & conservation

Biotechnology

Novel enzyme discovery (bioprospecting)
Bioactive compound mining
Bioremediation of pollutants

Food & Fermentation

Fermented food safety (dairy, beer, sauerkraut)
Food spoilage detection
Probiotic strain characterisation

Agriculture

Plant-growth-promoting bacteria
Soil carbon cycling & nutrient availability
Crop protection & disease suppression

Epidemiology

Outbreak surveillance & pathogen detection
Antibiotic resistance gene tracking
Zoonotic spillover monitoring

Approaches in Metagenomics

Amplicon Metagenomics

(Targeted)

PCR-amplify a marker gene (16S, 18S, ITS) from the community, then sequence the amplicons.

✓ Advantages

- Low cost per sample
- Deep coverage of marker gene
- Well-established pipelines
- Handles complex communities

✗ Limitations

- Only reveals who is there (taxonomy)
- PCR bias and chimera formation
- No functional information
- Primer mismatches miss taxa

THIS COURSE FOCUS →

Shotgun Metagenomics

(Whole-genome)

Randomly fragment and sequence all DNA in the sample, no amplification step.

✓ Advantages

- Taxonomic + functional data
- No PCR bias
- Detect novel organisms
- Reconstruct genomes (MAGs)

✗ Limitations

- High cost & computation
- Host DNA contamination
- Requires large databases
- Rare taxa easily missed

Metatranscriptomics

(RNA-based)

Sequence RNA (mRNA) from the community to capture active gene expression.

✓ Advantages

- Shows what is actively expressed
- Captures community response
- Reveals metabolic activity

✗ Limitations

- RNA unstable — fast processing needed
- Complex analysis
- rRNA dominates (~95%)

SECTION 2

The 16S rRNA Gene

Structure • Variable Regions • Why It Matters

The 16S rRNA Gene

The 16S rRNA gene encodes the small subunit of the prokaryotic ribosome. It is ~1,540 bp long and contains:

9 Conserved Regions

Shared across all prokaryotes — used for universal primer design

9 Variable Regions (V1–V9)

Evolve at different rates — provide species/genus-level resolution

~1,540 bp total

Full-length sequenced with PacBio/ONT; V3-V4 (~460 bp) most common with Illumina

Universal marker

Present in all bacteria and archaea; absent from eukaryotes (use 18S/ITS instead)

16S rRNA Gene Variable Regions

Region	Forward Primer	Reverse Primer	Read Length
V4	515F (GTGYCAGCMGCCGCGGTAA)	806R (GGACTACNVGGGTWTCTAAT)	Illumina 2×250
V3-V4	341F (CCTACGGGNGGCWGCAG)	806R (GGACTACNVGGGTWTCTAAT)	Illumina 2×300
V1-V3	27F (AGAGTTTGATCMTGGCTCAG)	518R (ATTACCGCGGCTGCTGG)	Illumina 2×300
Full-length	27F (AGAGTTTGATCMTGGCTCAG)	1492R (TACGGYTACCTTGTTACGACTT)	ONT / PacBio

Primer choice determines which microbial groups are detected and at what taxonomic resolution.

Sequencing Platforms for Metagenomics

Illumina (Short Reads)

Specifications

- Read length: 2×150 or 2×300 bp
- Throughput: up to 3 Tb/run (NovaSeq)
- Error rate: <0.1% (very accurate)
- Cost: ~$50–200 per sample

Common Uses

- V4/V3-V4 amplicon sequencing
- Shotgun WGS (most common)
- Large cohort studies

Tools: FASTQC, Trimmomatic, fastp, QIIME2, DADA2

Oxford Nanopore (ONT)

Specifications

- Read length: up to >1 Mb (avg 5–50 kb)
- Throughput: 50–300 Gb/run (PromethION)
- Error rate: ~5% (improving with R10 chemistry)
- Cost: ~$500–800/flow cell (~$50–150/sample multiplexed)

Common Uses

- Full-length 16S sequencing
- Rapid pathogen detection
- Direct RNA sequencing

Tools: wf-16S (EPI2ME), EMU, guppy/dorado, minimap2

PacBio HiFi

Specifications

- Read length: 10–25 kb (HiFi)
- Accuracy: >99.9% (CCS mode)
- Lower throughput than Illumina
- Higher cost per Gb

Common Uses

- Full-length 16S/18S (near-perfect accuracy)
- Genome assembly (hybrid approach)
- Large structural variation

Tools: pbmm2, DADA2 (with HiFi reads)

Amplicon vs Shotgun vs Long-Read: When to Use Each

	Amplicon (Short-Read)	Shotgun WGS (Short-Read)	Long-Read (ONT / PacBio)
What is sequenced	16S / ITS / 18S amplicon	All DNA in sample (WGS)	Full-length amplicon or WGS
Cost per sample	~$50–100 (cheap)	~$100–300 (moderate–high)	~$50–150 ONT (falling fast)
Resolution	Genus/species (V4); species (FL)	Strain-level; functional genes	Species-level (FL 16S); strain (WGS)
Error profile	<0.1% substitution (Illumina)	<0.1% substitution (Illumina)	~5% indel-rich (ONT); <0.01% (PacBio HiFi)
Functional info	None (marker gene only)	Full metabolic potential + AMR	Full (WGS); none (amplicon)
Best for	Community profiling, large cohorts	Assembly, MAGs, functional analysis	Species ID in clinical; complex assembly
Key tools	QIIME2, DADA2, EMU	Kraken2, MEGAHIT, MetaBAT2	Dorado, EMU, wf-16S, metaFlye

Primer Choice for Amplicon Sequencing

16S rRNA — V4

V4 (~253 bp) · 515F / 806R

✓ Most widely used ✓ Illumina 2×150/250 ✓ EMP standard ✓ Good archaea coverage

✗ Misses some Thaumarchaeota ✗ Low resolution below genus level

16S rRNA — V1-V3

V1-V3 (~500 bp) · 27F / 518R

✓ This course's dataset (PRJNA683885) ✓ Longer amplicon → more resolution ✓ Classic legacy marker

✗ No standard pre-trained SILVA classifier — must build locally ✗ Requires 2×300 bp reads

16S rRNA — V3-V4

V3-V4 (~460 bp) · 341F / 806R

✓ Better taxonomic resolution ✓ Common in human microbiome studies ✓ SILVA v138.2 classifiers

✗ Requires 2×300 bp reads ✗ More PCR cycles → more chimeras

16S rRNA — Full-length

V1-V9 (~1500 bp)

27F / 1492R

✓ Species-level resolution ✓ ONT & PacBio HiFi only ✓ EMU & wf-16S designed for this

✗ Long reads required (ONT/PacBio) ✗ Higher error rate (ONT) ✗ Cannot use Illumina 2×300

ITS (Fungi)

ITS1 or ITS2 (~300 bp)

ITS1F / ITS2 or ITS3F / ITS4

✓ Standard for mycobiome studies ✓ UNITE v10.0 database ✓ High species-level resolution

✗ ITS length variability complicates DADA2 ✗ Consider ITSxpress for trimming

Sequencing Error Profiles & Quality Control

Illumina (MiSeq/NovaSeq)

<0.1%

Error type: Substitution errors

Quality drops at 3' end; systematic run-specific errors

QC/handling: DADA2 error model per run; fastp Q20 filter

Oxford Nanopore (R10)

~5%

Error type: Indel errors (ins/del)

Homopolymer runs (poly-A/T stretches) are hardest; R10 much better than R9

QC/handling: EMU / minimap2 handle indels; Dorado recalibration

PacBio HiFi (CCS)

<0.01%

Error type: Substitution errors

Multiple passes produce near-perfect consensus; lowest error of any long-read

QC/handling: DADA2 with HiFi-specific parameters; pbmm2

SECTION 3

Reference Databases

SILVA • Greengenes2 • NCBI RefSeq • UNITE • GTDB

Key Reference Databases

SILVA

v138.2

16S / 18S / 23S rRNA

Gold standard rRNA database (July 2024). Curated alignments for all three domains of life.
Over 9.4 million SSU sequences. QIIME2 classifiers trained on SILVA are widely used.

Greengenes2

2024.09

16S rRNA

Rebuilt integrating WoL2 (Web of Life 2), a phylogenomic reference tree from ~10,000+ genomes.
Better species-level resolution. Default for recent QIIME2 tutorials.

NCBI RefSeq

Live

Full genomes

Comprehensive repository of reference sequences.
Used for Kraken2 database, BLAST searches, and downloading reference genomes.

UNITE

10.0

ITS (Fungi)

Species Hypothesis (SH) based fungal database. Essential for amplicon studies targeting ITS1/ITS2 regions for mycobiome analysis.

GTDB

r220

Whole genomes

Genome Taxonomy Database, phylogenomics-based prokaryote taxonomy (April 2024 release). Standardised nomenclature used with GTDB-Tk for MAG classification.

eggNOG 5.0

5.0.2

Proteins / COG

Hierarchical orthologous groups.
Used by eggNOG-mapper for rapid functional annotation. Maps genes to COG categories, GO terms and KEGG pathways.

SECTION 4

The Amplicon Analysis Pipeline

From raw reads to taxonomy and feature tables

Amplicon Analysis Workflow

01 Raw Reads

FASTQC + MultiQC

Quality control of raw FASTQ files — check quality scores, adapter content, GC bias.

02 Trimming & Filtering

Trimmomatic / fastp

Remove low-quality bases, adapters; filter very short reads.

03 Import to QIIME2

qiime tools import

Convert FASTQ files to QIIME2 artifact format (.qza). Define manifest file.

04 Denoise & Chimera Removal

DADA2 / Deblur

Error correction and chimera removal; generate ASVs (Amplicon Sequence Variants). DADA2 models sequencing errors per run.

05 Taxonomy Assignment

q2-feature-classifier

Classify ASVs using SILVA or Greengenes2 reference database.

06 Diversity & Visualisation

q2-diversity / R

Alpha/beta diversity, taxonomy bar plots, statistical testing.

🔬 **Long-Read Track (ONT):** Basecall (Dorado) → Full-length 16S reads → EMU (species-level abundance) or wf-16S (EPI2ME Nextflow workflow) | Skips QIIME2 DADA2 — uses alignment-based error handling instead

Oxford Nanopore 16S Tools

For long-read ONT 16S sequencing, different pipelines are used compared to Illumina:

EMU

Exact Mapping Utilities

Species-level 16S abundance estimation for ONT reads. Uses multi-mapping alignment to handle errors. Supports full-length 16S.

Pipeline steps:

1. Input: raw FASTQ from ONT
2. Maps reads to EMU default database (SILVA-based)
3. Outputs: species-level abundance table
4. Compatible with QIIME2 via q2-emu plugin

wf-16S

EPI2ME Nextflow workflow

Official Oxford Nanopore Technologies workflow for 16S/18S amplicon analysis. Nextflow-based, portable, cloud-ready.

Pipeline steps:

1. Input: FASTQ or POD5 files
2. Basecalling → Demultiplexing → Classification
3. Uses Minimap2 + SILVA/Greengenes2
4. Outputs: abundance table + report

Dorado

Basecaller (ONT)

GPU-accelerated basecaller from ONT. Converts raw signal (POD5) to FASTQ. Supports demultiplexing and modified base detection.

Pipeline steps:

1. Input: POD5 raw signal files
2. Basecalling model selection (fast/hac/sup)
3. Output: FASTQ with quality scores
4. Replaces older Guppy basecaller

Tools We Use in This Course

Day 1–2 Amplicon & Diversity	FASTQC · MultiQC · SEQKIT · QIIME2 · DADA2 · EMU · epi2me/wf-16S · phyloseq · vegan · ggplot2
Day 3 Shotgun & Assembly	fastp · SEQKIT · Bowtie2 · BWA · Samtools · Kraken2 · Bracken · MetaPhlAn4 · MEGAHIT · SPAdes · QUAST
Day 4 MAGs	MetaBAT2 · DAS_Tool · CheckM2 · GTDB-Tk · Bowtie2 · Samtools
Day 5 Functional Analysis	Prodigal · Prokka · eggNOG-mapper · HUMAnN3 · R (pathway plots)

All tools available via module load. See environment/metagenomics.yml and environment/qiime2.yml for installation.

AFTERNOON PRACTICAL

Today's Hands-On Session

1 QC Your Reads

Run FASTQC on raw gut microbiome 16S reads. Generate a MultiQC summary report. Inspect quality plots and identify potential issues.

2 Explore with SEQKIT

Use seqkit stats to get read count, length distributions, and GC content. Practice subsetting and filtering with seqkit grep, head, sample.

3 QIIME2 Amplicon Workflow

Import paired-end reads → run DADA2 denoising → assign taxonomy with SILVA classifier → generate taxonomy bar plots and feature tables.

4 ONT 16S with EMU (Bonus)

If time allows, process a small ONT 16S dataset using EMU. Compare taxonomy profiles to the Illumina-based results from step 3.

Day 1 Summary

Metagenomics	Culture-independent study of genetic material from complex environmental communities.
Amplicon sequencing	PCR-amplifies a marker gene (16S rRNA) for community profiling — cost-effective and well-established.
16S rRNA gene	~1,540 bp; 9 variable regions; V4 most common for Illumina; full-length for ONT/PacBio.
Databases	SILVA & Greengenes2 for 16S taxonomy; NCBI for reference genomes; GTDB for standardised taxonomy.
Pipeline	FASTQC → QIIME2 import → DADA2 (ASVs) → taxonomy classification → diversity analysis.
ONT 16S	Full-length 16S with EMU or wf-16S provides species-level resolution not achievable with short reads.

Day 2 tomorrow → Diversity Analysis (alpha diversity, beta diversity, ordination, statistics)

KEY REFERENCES

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Callahan et al. (2016)

DADA2: High-resolution sample inference from Illumina amplicon data

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Greengenes2 unifies microbial data in a single reference tree

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Emu: species-level profiling of full-length 16S rRNA ONT data

Nature Methods 19: 845–853

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SeqKit: a cross-platform ultrafast toolkit for FASTA/Q manipulation

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